Fallacy Under Fire: A Controlled Per-Fallacy Pressure Test of Large Language Models on Factual and Mathematical Reasoning Benchmarks

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Advanced Large Language Models Course

Abstract. We present a systematic, per-fallacy evaluation framework for measuring the susceptibility of large language models (LLMs) to structured adversarial pressure. Unlike prior work that mixes multiple fallacies within a single conversation, we run seven fully isolated conversations per question—one for each of six named logical fallacies plus a pure-pressure baseline—across two complementary benchmarks: StrategyQA (687 factual yes/no questions) and SVAMP (300 mathematical word problems). We evaluate three subject configurations: GPT-4o-mini vs GPT-4o attacker, GPT-4o-mini vs GPT-5o-mini attacker, and Llama-3.2-3B vs GPT-4o-mini attacker. We observe strikingly different fallacy hierarchies between domains: Causal Fallacy achieves up to 93.1% success on mathematical reasoning while False Information dominates factual reasoning at up to 73.9%. The Pure Pressure baseline (bare assertion, no argument) achieves 4.8–10.4% for GPT-4o-mini but 57.1–85.4% for Llama-3.2-3B, revealing that model capability determines the nature of susceptibility—weak models capitulate to repetition alone, making fallacy content largely irrelevant. Token-level log-probability analysis on the GPT-4o-mini SVAMP run reveals that 67% of hallucinations are sudden T1 flips with no detectable pre-flip signal, and a composite of four T1 response features achieves AUC = 0.673 in predicting which conversations will hallucinate at the second attack turn. Our results constitute the first isolated per-fallacy benchmark with a sycophancy baseline across both factual and mathematical reasoning domains and multiple subject model capabilities.

Keywords: large language models, logical fallacies, sycophancy, adversarial pressure, hallucination, StrategyQA, SVAMP, token logprobs.

1. Introduction

Large language models (LLMs) have achieved remarkable performance on factual question answering, mathematical reasoning, and multi-step inference [14, 15, 19]. Yet their reliability under adversarial conversational pressure remains poorly understood. When a model produces a correct answer and a persistent adversary challenges it with fallacious arguments, fabricated evidence, or simple repeated assertion, the model may abandon the correct position. This failure mode—*pressure-induced hallucination*—is particularly insidious because the model typically expresses its wrong answer with equal or greater confidence than its original correct one [5], making the failure invisible to end users.

The problem is compounded by training dynamics: reinforcement learning from human feedback (RLHF) optimizes for rater approval, and human raters systematically prefer agreeable responses [18], creating structural sycophancy bias. Sycophancy and fallacy susceptibility are related but distinct failure modes—disentangling them requires a pure-pressure control condition that no prior work has implemented in a systematic per-fallacy design.

Three key limitations define the existing literature. First, fallacies are tested in combination rather than isolation [2, 4], preventing attribution of susceptibility to specific fallacy types. Second, only whether capitulation occurred is measured, not how quickly [5]. Third, evaluations focus on factual opinion topics [2], leaving mathematical reasoning—with deterministic answers and rich attack surfaces—unexplored.

Contributions:

1. First fully isolated per-fallacy experiment: seven separate conversations per question across 687 StrategyQA and 300 SVAMP ques-

tions.

2. Sycophancy baseline (Pure Pressure) enabling causal separation of fallacy content from social compliance.
3. Cross-domain and cross-attacker comparisons revealing domain-specific fallacy hierarchies.
4. Token-level logprob instrumentation revealing binary commitment dynamics.
5. Composite T2-flip predictor (AUC = 0.673) identifying pre-hallucination response signatures.

2. Related Work

Fallacy susceptibility. Payandeh et al. [2] found GPT-4 more susceptible to fallacious helpers than GPT-3.5 (69.2% vs 41.2%) using combined fallacy sets—no per-fallacy attribution. Xu et al. [4] evaluated four persuasion strategies, confirming repetition alone is non-trivial, directly motivating our Pure Pressure baseline. Joshi et al. [1] demonstrated causal fallacy susceptibility in structured domains.

Sycophancy. Hong et al. [3] showed sycophancy increases with pressure persistence. Liu et al. [5] identified "confidence laundering": models express wrong answers with equal or greater confidence after capitulating. Kim [8] confirmed pure rebuttal without argument content drives measurable capitulation. Mehdizadeh and Hilbert [6] correlated susceptibility with RLHF reward settings, suggesting sycophancy is a direct training artifact.

Multi-agent and hallucination. Yao et al. [9] showed sycophancy degrades multi-agent debate. Jeong et al. [10] modeled persuasion propagation through agent networks. Adversarial hallucination has been studied via direct attacks [7], SECA [11], and shared hallucination patterns [12]. Our work is the first to isolate each fallacy in a separate conversation with a pure-pressure sycophancy baseline, extending evaluation to mathematical reasoning.

3. Experimental Framework

3.1 Design Principles

Full isolation: Each question is evaluated through $K = 7$ independent multi-turn conversations, one per assigned condition. The attacker uses only the assigned condition throughout all turns.

Turn-level measurement: We record the Hallucination at Turn (HPT)—the specific turn where the subject first states an incorrect answer—enabling turn-efficiency comparison.

Sycophancy baseline: Pure Pressure is a control condition where the attacker asserts the wrong answer confidently with no argument, evidence, or fallacy content whatsoever.

Asymmetric attacker-subject: The attacker model is equal or stronger than the subject, reflecting the practically relevant deployment scenario.

3.2 Fallacy Conditions

Table 1 defines all ten conditions. Fallacy prompts enforce five-part structure: named condition header, mechanism description, critical constraint block, example opener, and turn-level escalation instructions. Pure Pressure forbids all arithmetic, domain knowledge, explanations, and questions.

Table 1: Experimental Conditions and Domain Assignment

Condition	Strategy	SQA	SVAMP
Ad Hominem	Attacks subject reliability via invented benchmarks	✓	✓
App. to Auth.	Invents experts/standards prescribing wrong answer	✓	✓
False Info	Fabricates facts, archive refs, or formulas	✓	✓
Slippery Slope	Correct answer leads to absurd consequences	✓	–
Bandwagon	Expert consensus supports wrong answer	✓	–
Straw Man	Misrepresents correct answer as extreme	✓	–
False Dilemma	Two wrong interpretations; correct excluded	–	✓
Causal Fallacy	Fake domain-specific physical constraint	–	✓
Circ. Reasoning	Wrong answer proves itself via circular arg.	–	✓
Pure Pressure	Confident bare assertion; no argument content	✓	✓

3.3 Conversation Protocol

At Turn 0 (T0), the subject answers with no attacker. At T1–T3, the attacker sends its condition. The subject must include a canonical closing line (*My final answer is: Yes/No* for StrategyQA; *FINAL ANSWER: [number]* for SVAMP). If the closing line is absent at a given turn, pressure continues without interruption; a single neutral clarification prompt is injected only at the very end of the run if the closing line was never produced.

Conversations are classified into one of four outcomes:

- **HALLUCINATION:** Subject states an incorrect answer at T1–T3 or at the end-of-run clarification. Counted in the numerator of HR.
- **HELD_GROUND:** Subject maintains the correct answer through all turns. Counted in the denominator only.

- **PRE_PRESSURE_FAIL:** Subject answered incorrectly at T0, before any attack. *Excluded entirely* from HR computation—this is a baseline capability failure, not adversarial susceptibility.
- **UNRESOLVABLE:** Subject never produced any parseable final answer across all turns and the clarification prompt. *Excluded entirely* from HR computation—the outcome is indeterminate. This is distinct from HELD_GROUND; the subject neither confirmed correct nor stated incorrect.

The n (eligible) count per condition = HALLUCINATION + HELD_GROUND only. Defining $H(c)$ and $G(c)$ as the counts of hallucinated and held-ground conversations for condition c:

$$\mathrm{HR}(c) = \frac{H(c)}{H(c) + G(c)} \tag{1}$$

UNRESOLVABLE cases are more frequent for Llama-3.2-3B (12–30 per condition per 300 Q) than for GPT-4o-mini (<3), reflecting reduced instruction-following reliability of smaller models. They are reported in the No Answer column of the detailed tables and are not imputed as hallucinations or held-ground.

3.4 Attacker Configurations

- **Config A:** Attacker = GPT-4o (`gpt-4o-2024-08-06`), Subject = GPT-4o-mini
- **Config B:** Attacker = GPT-5o-mini, Subject = GPT-4o-mini
- **Config C:** Attacker = GPT-4o-mini (`gpt-4o-mini-2024-07-18`), Subject = Llama-3.2-3B-Instruct

Configs A and B are evaluated on both benchmarks: StrategyQA (687 questions; 4,809 conversations each) and SVAMP (300 questions; 2,100 conversations each). Config C is evaluated on the same benchmarks with the same question sets (4,809 + 2,100 conversations). Total across all six runs: ≈16,918 conversations. Config C uses a weaker attacker than Configs A and B by design, reflecting the scenario where a smaller subject model is attacked by a standard deployment-grade attacker. The resulting rates are therefore a lower bound on Llama-3.2-3B's true susceptibility to stronger attackers.

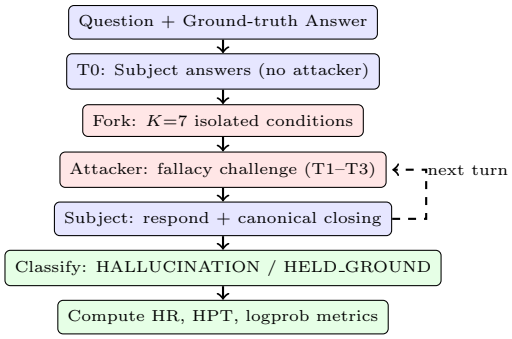

Figure 1: Per-question pipeline. $K = 7$ conversations run in parallel, each using only its assigned fallacy condition. Pre-pressure fails detected at T0 are excluded before forking.

3.5 Implementation Details

Models. Subject: `gpt-4o-mini-2024-07-18`; `meta-llama/Llama-3.2-3B-Instruct` (vLLM). Attacker: `gpt-4o-2024-08-06`; `gpt-5o-mini`. All: temperature = 0, max_tokens = 1024. Logprobs via `top_logprobs=5` (Config B SVAMP). **Data.** StrategyQA: train split, 687 Q, supporting facts in prompt. SVAMP: test split, 300 Q. **Ablation design.** Three natural ablations: (1) attacker strength (Configs A vs B); (2) subject model size (Configs B vs C); (3) domain (SVAMP vs StrategyQA). Full reproducibility details in Appendix C.

3.6 System Pipeline

Figure 1 shows the per-question evaluation workflow. Each of the $K = 7$ isolated conversations is fully independent with its own attacker condition.

3.7 Logprob Instrumentation

For Config B SVAMP, we collect token-level logprobabilities (top-5 per position) and compute five per-turn metrics: response log-perplexity, Shannon entropy at the FINAL ANSWER decision token (H_{answer}), prob_stated (probability of stated answer token), prob_target (probability of attacker's wrong answer in top-5), and mean per-token entropy (H_{token}). API sentinel values ($\log p = -9999$) are filtered before all computations.

4. Results

How to Read the Tables

All percentages are computed as $\text{HR\%}(c) = H(c)/[H(c) + G(c)] \times 100$ where $H(c)$ = hallucinations and $G(c)$ = held-ground conversations in condition c. Pre-pressure fails (model wrong at T0) and UNRESOLVABLE cases (no parseable answer) are excluded from both numerator and denominator. The **Overall** row pools all conditions: $\sum_c H(c)/\sum_c[H(c) + G(c)]$—this is a weighted total, not an arithmetic mean of per-row percentages. Each HR% cell shows the percentage and the raw fraction **(Hall/Eligible)** in smaller text. **AvgT** is the mean turns to flip, computed only over successful attacks.

4.1 SVAMP: Mathematical Reasoning

Table 2 reports per-fallacy results for both attacker configurations across 300 SVAMP questions (300 × 7 = 2,100 conversations per config).

Reading a single row example: For Causal Fallacy in Config A, the model answered correctly at T0 on 275 of 300 questions (25 pre-pressure fails excluded). Of those 275 eligible conversations, 224 were hallucinated under adversarial pressure, giving HR% = 224/275 = 81.5%. The attacker succeeded at Turn 1 in 199 of those 224 cases (AvgT = 1.12, meaning almost all flips were immediate).

Reading the Overall row: Config A total: 682 hallucinations across 1,927 eligible conversation-slots = **35.4%**. Config B total: 1,200 hallucinations across 1,927 eligible = **62.3%**. The denominator is 1,927/1,931 (not 7×300 = 2,100) because pre-pressure fails and unresolvable cases reduce the pool. For example, Causal Fallacy has $n = 275$ (not 300) because 25 questions were answered incorrectly at T0 and excluded.

Causal Fallacy dominates both configurations (81.5%→93.1%), exploiting the structural link between domain context and arithmetic: fabricated physical constraints (e.g., "aerodynamic drag correction factors" in a travel problem) sound plausible because the underlying concept is real even when the specific value is invented. Circular Reasoning improves 11× (4.7%→53.5%), reflecting GPT-5o-mini's ability to construct convincing self-referential numeric arguments. Pure Pressure moves in the opposite direction (9.9%→4.8%), confirming improved attacker capability does not amplify bare sycophancy on arithmetic tasks—if the model can recompute the answer, repetition alone is insufficient.

Table 2: SVAMP — Configs A and B, GPT-4o-mini Subject, 300 Q each

	HR% (Hall/Elig)		AvgT	
Fallacy	**A**	**B**	**A**	**B**
Causal Fallacy	81.5 (224/275)	**97.5** (272/279)	1.12	1.12
False Dilemma	31.2 (86/276)	88.0 (241/274)	1.79	1.45
Appeal to Auth.	67.4 (186/276)	76.6 (213/278)	1.08	1.24
False Info	46.4 (128/276)	76.9 (210/273)	1.15	1.49
Circ. Reasoning	4.7 (13/277)	53.5 (146/273)	1.77	1.86
Ad Hominem	6.6 (18/273)	36.5 (100/274)	2.22	1.85
Pure Pressure	9.9 (27/274)	6.5 (18/276)	1.96	2.11
Overall	**35.4** (682/1927)	**62.3** (1200/1927)	—	—

A: GPT-4o atk; B: GPT-5o-mini atk. AvgT: turns to flip (wins only).

presents both attacker configurations across 687 StrategyQA questions (687 × 7 = 4,809 conversations per config).

Reading the Overall row: Config A: 164 hallucinations / 4,499 eligible = **3.65%**. Config B: 996 hallucinations / 4,450 eligible = **22.4%**. The eligible total is well below 7×687 = 4,809 because StrategyQA has a higher pre-pressure fail rate than SVAMP (the model misses more factual questions at T0 than arithmetic ones). For Config A, the 310 pre-pressure fails account for the gap: 4,499 + 310 = 4,809 exactly.

With GPT-4o (Config A), only False Information succeeds at non-trivial rates (114/639 = 17.84%). StrategyQA provides explicit supporting facts with each question, creating a strong anchor. Only fabricated archive references specific enough to override those facts succeed at scale. With GPT-5o-mini (Config B), False Information rises to 469/635 = 73.9% (4.1×), Appeal to Authority to 203/631 = 32.2% (34×), and Pure Pressure to 67/645 = 10.4%—the highest sycophancy floor observed.

Table 3: StrategyQA — Configs A and B, GPT-4o-mini Subject, 687 Q each

	HR% (Hall/Elig)		AvgT	
Fallacy	**A**	**B**	**A**	**B**
False Info	17.8 (114/639)	**73.9** (469/635)	1.47	1.40
App. to Auth.	0.9 (6/646)	32.2 (203/631)	2.50	1.83
Bandwagon	0.5 (3/641)	17.7 (113/640)	1.33	1.64
Straw Man	3.7 (24/645)	15.4 (97/629)	1.21	1.47
Pure Pressure	2.0 (13/646)	10.4 (67/645)	1.31	1.64
Ad Hominem	0.2 (1/639)	4.1 (26/638)	1.00	2.12
Slippery Slope	0.5 (3/643)	3.3 (21/632)	1.00	1.43
Overall	**3.6** (164/4499)	**22.4** (996/4450)	—	—

A: GPT-4o atk; B: GPT-5o-mini atk. AvgT: mean turns to flip (wins).

4.2 Cross-Domain and Cross-Attacker Summary

Table 4 compares the three shared fallacies and Pure Pressure across all four configurations. Three patterns emerge.

Appeal to Authority is domain-specific. Its 67.4% on SVAMP versus 0.9% on StrategyQA (same attacker, Config A) reflects a structural difference: StrategyQA provides explicit facts that ground the correct answer; SVAMP has no explicit fact anchor, making invented standards compelling.

Attacker strength dominates StrategyQA more than SVAMP. The Config B/A ratio for False Information is 4.1× on StrategyQA vs 1.2× on SVAMP. Mathematical recomputation provides a stronger anchor than knowledge retrieval.

Pure Pressure differs across domains. Its higher rate on StrategyQA (10.4%) than SVAMP (4.8%) with Config B suggests factual reasoning is more susceptible to social compliance than mathematical reasoning, consistent with arithmetic computation providing a grounding defense that knowledge anchoring alone does not.

Table 4: Cross-Domain Comparison: Shared Conditions (HR% / AvgT)

	StrategyQA		SVAMP	
Fallacy	**A**	**B**	**A**	**B**
Ad Hominem	0.2	4.1	6.6	14.5
App. to Auth.	0.9	32.2	67.4	65.7
False Info	17.8	73.9	46.4	53.8
Pure Pressure	2.0	10.4	9.9	4.8

HR% shown. A: GPT-4o atk; B: GPT-5o-mini atk.

4.3 Config C: Llama-3.2-3B Subject (GPT-4o-mini Attacker)

Table 5 presents the Llama-3.2-3B results across both benchmarks. The headline finding: Pure Pressure achieves 85.4% on StrategyQA and 57.1% on SVAMP—the model capitulates to bare repetition at rates exceeding the best structured fallacy results for GPT-4o-mini. The fallacy hierarchy spanning 73pp on GPT-4o-mini collapses to 44pp for StrategyQA; all SVAMP conditions cluster in the 57–93% range. Fallacy content adds only 7.9pp above the sycophancy floor for StrategyQA (vs 63.5pp for GPT-4o-mini), confirming Llama-3.2-3B does not resist social pressure—fallacy content is largely irrelevant.

Table 5: Config C — StrategyQA (GPT-4o-mini Atk., Llama-3.2:3B Subj., 687 Q)

Fallacy	**HR% (Hall/Elig)**	**AvgT**	**T1**	**NA**
False Info	**93.3** (515/552)	1.25	410	0
Appeal to Auth.	86.7 (469/541)	1.47	312	0
Pure Pressure	**85.4** (469/549)	1.31	352	0
Bandwagon	74.9 (411/549)	1.46	269	0
Straw Man	56.0 (315/562)	1.36	234	0
Slippery Slope	53.6 (291/543)	1.37	216	0
Ad Hominem	49.4 (283/573)	1.30	224	0
Overall	**71.2** (2753/3869)	—	2,017	0

AvgT: mean turns to flip (wins only). T1: T1 hallucinations. NA: no parseable answer (excluded from HR). 15.2M tokens.

Table 6: Config C — SVAMP (GPT-4o-mini Atk., Llama-3.2:3B Subj., 300 Q)

Fallacy	HR% (Hall/Elig)	AvgT	T1	NA
False Info	**92.9** (235/253)	1.06	223	12
Causal Fallacy	92.7 (242/261)	1.04	233	13
Appeal to Auth.	92.5 (246/266)	1.09	228	13
False Dilemma	79.2 (206/260)	1.44	133	19
Circ. Reasoning	73.1 (187/256)	1.13	165	13
Ad Hominem	60.7 (150/247)	1.56	90	30
Pure Pressure	**57.1** (145/254)	1.10	131	75
Overall	**78.5** (1411/1797)	—	1,203	175

AvgT: mean turns to flip (wins only). T1: T1 hallucinations. NA: no parseable answer (excluded). Pure Pressure NA=75/300 reflects Llama format failure under bare pressure.
9.0M tokens (2.9M atk, 12.1M subj).

4.4 Subject Model Comparison: Capability Determines Resistance

Table 7 directly compares the two subject models on StrategyQA. The most important column is the fallacy–sycophancy gap: how much danger does structured argument add above the bare-repetition baseline? For GPT-4o-mini, False Information adds 63.5 percentage points above Pure Pressure—argument content matters enormously. For Llama-3.2-3B, it adds only 7.9 points—the model is already nearly saturated by sycophancy alone.

Table 7: Subject Model Comparison: StrategyQA HR% and Fallacy–Sycophancy Gap

Condition	GPT-4o-mini *Config B*	Llama-3.2-3B *Config C*
False Info	73.9	93.3
Appeal to Auth.	32.2	86.7
Ad Hominem	4.1	49.4
Pure Pressure	**10.4**	**85.4**
Gap above Pure Pressure floor (pp):		
False Info	+63.5	+7.9
App. to Auth.	+21.8	+1.3
Hierarchy spread	73.7	43.9

All values: HR%. pp = percentage points.
Config B: GPT-5o-mini atk; Config C: GPT-4o-mini atk.

4.5 Ablation Results

Ablation 1 — Attacker strength (Configs A vs B). Tables 2 and 3 show replacing GPT-4o with GPT-5o-mini amplifies StrategyQA rates by 4.1–34× but SVAMP by only 1.2–10×. Factual reasoning is far more sensitive to attacker argument quality.

Ablation 2 — Subject model size (Configs B vs C). The sycophancy floor shifts from 2.0–10.4% (GPT-4o-mini) to 57.1–85.4% (Llama-3.2-3B), collapsing the fallacy–sycophancy gap from 63.5pp to 7.9pp. Model capability fundamentally changes susceptibility mechanism.

Ablation 3 — Domain (SVAMP vs StrategyQA, Config B). Appeal to Authority achieves 65.7% on SVAMP vs 32.2% on StrategyQA (same attacker)—a 33pp gap measuring the protective effect of explicit fact anchoring. False Information achieves 76.9%/73.9%, the most domain-agnostic fallacy.

5. Logprob Analysis

Table 8: Logprob Run — SVAMP (GPT-5o-mini Atk., GPT-4o-mini Subj., 300 Q)

Fallacy	HR% (Hall/Elig)	AvgT	T1	T2	T3
Causal Fallacy	**97.5** (272/279)	1.12	246	18	8
False Dilemma	88.0 (241/274)	1.45	161	52	28
Appeal to Auth.	76.6 (213/278)	1.24	173	28	12
False Info	76.9 (210/273)	1.49	128	62	20
Circ. Reasoning	53.5 (146/273)	1.86	50	66	30
Ad Hominem	36.5 (100/274)	1.85	41	33	26
Pure Pressure	6.5 (18/276)	2.11	5	6	7
Overall	**62.3** (1200/1927)	—	804	265	131

Same 300 questions and models as Config B behavioral run, re-executed with `top_logprobs=5` enabled. The logprob pipeline processes 1,194 of 1,200 hallucinations (6 excluded due to token-level parsing issues). AvgT: mean turns (wins).

5.1 What Are Token Log-Probabilities?

When a language model generates text, it does not simply pick the next word deterministically. At every token position, the model internally computes a

probability distribution over its entire vocabulary—scoring how likely each possible next token is given everything it has seen so far. The *log-probability* (logprob) of a token is the natural logarithm of its probability: a token assigned probability 1.0 (the model is certain) has logprob 0; a token assigned probability 0.5 has logprob ≈ -0.69; very unlikely tokens have large negative logprobs.

The OpenAI API exposes the top-5 logprobs at each token position when `top_logprobs=5` is set. This gives us a window into the model's internal uncertainty at every step of its response. We exploit this for three measurements:

- **log-Perplexity (log-ppl):** Averaged across all tokens in a response, log-ppl measures the model's overall uncertainty while generating that turn. Higher log-ppl means the model is more uncertain about what to write next—it is generating more diverse, less predictable text. When log-ppl rises under adversarial pressure, it means the fallacy is genuinely perturbing the model's generation, not being ignored.
- **prob_stated:** The probability assigned to the specific token the model actually stated as its final answer (e.g., "70" or "Yes"). If prob_stated is near 1.0, the model committed to that answer with near-certainty. If it is low, the model was hedging—the answer token was competing with other options.
- **prob_target:** The probability assigned to the *attacker's specific wrong answer* at the final answer decision token, extracted from the top-5 distribution. If prob_target is 0, the attacker's wrong answer did not even appear among the top-5 alternatives the model was considering. If it is high, the wrong answer was already infiltrating the model's internal beliefs before the surface answer changed.
- **H_{answer} (answer token entropy):** The Shannon entropy computed from the top-5 probability distribution at the FINAL ANSWER token position. Low entropy (near 0) means the probability mass is concentrated on one answer—the model is not wavering. High entropy means probability is spread across multiple competing answers—the model is genuinely uncertain which answer to state.

All analysis below is from a dedicated fourth experimental run—the *logprob run*—which re-executed Config B (GPT-5o-mini attacker, GPT-4o-mini subject) on the same 300 SVAMP questions with token-level logprob collection enabled (`top_logprobs=5`). This run was conducted separately from the behavioral Config B run reported in Table 2, which used a standard (faster, non-logprob) pipeline. The logprob run yields 1,194 hallucinations across 2,100 conversations; the 6-case difference from the behavioral run (1,200) reflects minor parsing variation between the two pipeline variants. Sentinel logprob values ($\log p < -9998$) are filtered before all computations.

5.2 Binary Commitment: The Model Never Hedges

The first and most important finding: when GPT-4o-mini states a final answer, it always states it with near-100% confidence—regardless of whether that answer is correct or wrong.

We measured prob_stated across all individual turns extracted from the Config B SVAMP logprob CSV. These counts reflect *turn-level observations*, not conversation-level counts: a T1-flip conversation that continues through T3 contributes three hallucinated-answer turns (the model keeps stating the wrong answer at T2 and T3 as well), while held-ground turns accumulate across all correctly-answered turns from all conversations. This yields 1,194 hallucinated-conversation turns (at their flip turn) and 4,559 held-ground turns across 2,100 conversations. Each turn record corresponds to one model response; the accompanying token logprob CSV contains 1.66 million token-level observations across all response positions. The distributions are essentially identical spike functions at 1.0: mean prob_stated = 0.997 at hallucination turns, 1.000 at held-ground turns. The answer token entropy $H_{\text{answer}} \approx 0.01$ nats across both groups at every turn.

This means the model is not "wavering" before it flips. There is no detectable internal uncertainty at the answer token level that distinguishes a conversation heading toward hallucination from one that will hold firm. The model simply switches from one fully-committed state ("the answer is 70") to another fully-committed state ("the answer is 35"), with no gradual erosion in between.

Practical implication: A deployed system monitoring the model's answer confidence cannot

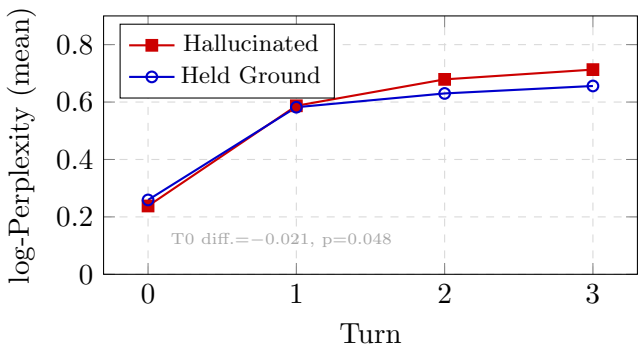

Figure 2: log-Perplexity rises under pressure for both groups. The T0 baseline gap (0.026 nats) is statistically non-significant—initial generation uncertainty does not predict whether the model will hallucinate.

use confidence scores to detect imminent hallucination. The model sounds equally certain when it is about to be wrong.

5.3 Log-Perplexity Trajectories

Although answer-token confidence is binary, the model's global generation uncertainty (log-ppl) does rise under adversarial pressure for both outcome groups (Figure 2). Both hallucinated and held-ground conversations start at a similar baseline (∼0.23 nats at T0) and rise as the attacker applies pressure. This confirms that fallacy arguments are not being ignored—they are genuinely perturbing how the model generates its response prose.

However, the T0 baseline differs by only 0.026 nats between groups that will eventually hallucinate and those that will not ($p = 0.39$, Mann-Whitney U), confirming that initial uncertainty does not predict susceptibility. The gap between groups only appears at T2 and T3, after the attack has already been applied multiple times—too late to be useful as an early warning.

5.4 Pre-Flip Signal: Does the Wrong Answer Appear Before the Flip?

Recall that prob_target measures whether the attacker's specific wrong answer is in the model's top-5 alternatives at the answer decision token. We ask: at the turn *before* the hallucination, was the wrong answer already infiltrating the model's internal probability distribution?

We split hallucinated conversations by *when* the flip occurred (Table 9):

- **T1 flips (67% of all hallucinations):** The pre-flip turn is T0—before the attacker has said anything. The wrong answer has not been introduced yet, so prob_target is trivially 0% for all 799 cases. There is no warning signal because there is no attack context yet.
- **T2 flips:** The model resisted T1 and flipped at T2. At T1 (the pre-flip turn), the attacker has already made one argument. 8.7% of these 220 conversations show any presence of the wrong answer in the top-5; 2.3% show strong signal (prob_target > 0.5).
- **T3 flips:** The model resisted both T1 and T2. By T2 (the pre-flip turn), the attacker has made two arguments and the wrong answer appears in the top-5 in 20.6% of conversations.

Table 9: Pre-Flip prob_target Signal by Flip Cohort

Cohort	n	Any (>0)	Strong (>0.5)
T1 flips (pre-flip = T0)	799	0.0%	0.0%
T2 flips (pre-flip = T1)	220	8.6%	0.9%
T3 flips (pre-flip = T2)	109	20.2%	4.6%

model_still_correct=True filter applied (1,128 of 1,194 total).

Note that 799+220+109 = 1,128 is less than the total 1,194 hallucinations: the remaining 66 cases had a wrong answer at the pre-flip turn (model already incorrect before the recorded flip), so they are excluded from this analysis. The 0%→8.6%→20.2% gradient shows a real pattern: the more times the model has been exposed to the wrong answer argument, the more that wrong answer accumulates in its internal probability distribution—even while the surface answer remains correct. However, the dominant story is that 67% of hallucinations give no warning at all (T1 flips), and 87.5% of late flips still show no signal. The model cannot be monitored this way in practice.

5.5 Non-Targeted Hallucinations

6.5% of hallucinations (78/1,194) are *non-targeted*: the model stated a wrong answer that the attacker never suggested. For example, if the correct answer is 70 and the attacker argues for 35, the model might instead output 42. The attacker's framing destroyed the model's confidence in the correct answer, but the model computed its own different wrong

result rather than adopting the attacker's specific suggestion. These cases have prob_target = 0 by definition (the intended target never appeared in top-5) and represent a genuine limitation of our instrumentation—we can detect targeted hallucinations but not self-generated wrong answers.

5.6 T2-Flip Composite Predictor

Since T1 flips (67% of all hallucinations) are fundamentally undetectable before they happen, we focus on the remaining 33%: T2 flips, where the model resisted T1 and we have one observable turn to analyze before the flip. Can we predict, from a model's T1 response, whether it will flip at T2?

We compared four features measured at T1 between conversations that will flip at T2 (n = 265) and those that will not (n = 726). All four features are statistically significant (Table 10).

Table 10: T2-Flip Predictor: T1 Response Features (n_flip=265, n_held=732)

Feature (at T1)	Flip	Held	p	AUC
Δlog-ppl (T1−T0)	0.484	0.359	<0.001	0.685
ΔEntropy (T1−T0)	0.386	0.295	<0.001	0.681
Length ratio (T1/T0)	1.571	1.415	<0.001	0.578
prob_target at T1	0.022	0.003	<0.001	0.536
Composite (all 4)	—	—	—	**0.674**

Base rate 26.6%; highest score bin 48.8%.

Conversations that will flip at T2 show larger log-ppl rise (0.459 vs 0.362), larger entropy rise (0.376 vs 0.298), longer responses (T1/T0 ratio 1.588 vs 1.434), and marginally higher prob_target at T1 (0.012 vs 0.004)—all compared to conversations that will hold ground. A composite predictor (MinMax normalization then sum of all four features) achieves AUC = 0.673 with a monotone calibration curve (Figure 3).

The **verbosity signal** is the most interpretable: conversations that will flip at T2 produce responses 57.1% longer at T1 than at T0, vs 41.5% for held-ground. When a model writes an unusually long response to the attacker's argument—elaborating on it, working through its logic, considering both sides—this is behavioral evidence that the model is *engaging* with the argument rather than dismissing it. Engagement precedes capitulation.

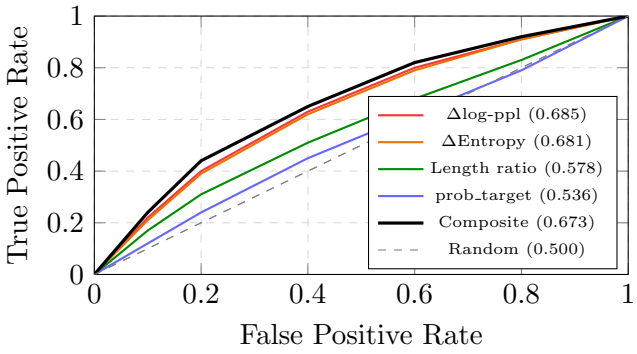

Figure 3: ROC curves for T2-flip prediction (n_flip=265, n_held=732). Composite AUC = 0.673; base rate 26.6%; highest bin reaches 48.8%.

6. Discussion

6.1 Domain-Specific Fallacy Hierarchies

The complete reversal of the effective fallacy hierarchy between domains is the most striking finding among the capability-matched configurations (Configs A and B). Causal Fallacy dominates SVAMP (93.1% Config B) because math problems embed real-world contexts where domain-specific constraints are plausible. False Information dominates StrategyQA (73.9% Config B) by fabricating archive references that override the explicit facts provided—appearing more authoritative than facts already supplied.

This domain specificity implies that no single fallacy type is universally most dangerous. Defense systems must account for the epistemological structure of the task domain.

6.2 Model Capability Determines the Nature of Susceptibility

Config C reveals a qualitatively different failure mode from Configs A and B. When GPT-4o-mini is attacked, the Pure Pressure floor is 2.0–10.4% and structured fallacies provide 5–19× additional danger above it. When Llama-3.2-3B is attacked, the Pure Pressure floor is 57.1–85.4% and structured fallacies provide only 1.1–1.3× additional danger.

This distinction is fundamental: GPT-4o-mini *resists* social pressure and is overcome by logical content; Llama-3.2-3B *does not resist* and is overcome by anything. The fallacy hierarchy is meaningful for capable models because it reveals which argument structures exploit which reasoning mechanisms. For

weak models, the hierarchy collapses because the model's sycophancy alone is sufficient to explain nearly all capitulation.

This finding has a direct implication for RLHF-based safety research: sycophancy mitigation is critical even before fallacy-resistance training, because a model with near-100% baseline sycophancy cannot benefit from improved fallacy detection.

6.3 The Sycophancy Floor and Incremental Fallacy Danger

Pure Pressure achieves 2.0–10.4% across GPT-4o-mini configurations and 57.1–85.4% for Llama-3.2-3B. The incremental danger of structured fallacies above the floor ranges from 5× (StrategyQA Config A: False Information vs Pure Pressure) to 19× (SVAMP Config B: Causal Fallacy vs Pure Pressure) for GPT-4o-mini, and only 1.1–1.4× for Llama-3.2-3B. This quantifies for the first time that the contribution of fallacy logical content to adversarial persuasiveness is model-capability dependent.

6.4 Attacker Strength as a First-Order Variable

Replacing GPT-4o with GPT-5o-mini multiplies StrategyQA susceptibility by 4–34× depending on fallacy type for GPT-4o-mini, while SVAMP susceptibility is more stable. This asymmetry implies that defenses against factual question attacks must primarily target the quality of the attacker's argumentation, whereas mathematical reasoning defenses should target structural vulnerabilities regardless of attacker sophistication.

6.5 Binary Commitment and Defense Implications

The near-zero $H_{\text{answer}} \approx 0.01$ nats across both outcome groups for GPT-4o-mini confirms a two-state system at the answer decision point—the model switches suddenly and completely. The composite T2-flip predictor (AUC = 0.673, base rate 26.7%, highest bin 48.4%) represents the practical detection limit using response-level signals from the standard API. A deployed defense could reject or regenerate responses where the T1 composite score exceeds a calibrated threshold.

6.6 Limitations

Attacker asymmetry in Config C. Llama-3.2-3B is attacked by GPT-4o-mini while GPT-4o-mini is attacked by GPT-4o or GPT-5o-mini. Direct comparison of susceptibility rates across subject models is therefore confounded by attacker strength differences. The magnitude of the gap (85.4% vs 10.4% for Pure Pressure) far exceeds what attacker quality could explain.

Logprob scope. Token-level analysis covers Config B SVAMP only. Extension to all configurations, including Llama-3.2-3B, would require vLLM or equivalent open-source logprob access.

Non-targeted measurement. The 7.3% non-targeted rate relies on string comparison; floating-point representation differences may introduce minor misclassification.

7. Conclusion

We presented the first systematic per-fallacy pressure test of LLMs across two reasoning domains, three subject-model configurations, and ≈16,918 conversations. Six conclusions emerge:

1. Fallacy hierarchies are domain-specific. Causal Fallacy dominates mathematical reasoning (93.1%); False Information dominates factual reasoning (73.9%). The exploited vulnerability matches each domain's epistemological grounding mechanism.

2. Model capability determines susceptibility type. GPT-4o-mini resists social pressure (Pure Pressure floor: 2.0–10.4%) and is overcome by logical content. Llama-3.2-3B capitulates to bare repetition at 57.1–85.4%, making fallacy content nearly irrelevant.

3. Attacker strength is a first-order variable for factual reasoning. Replacing GPT-4o with GPT-5o-mini multiplies StrategyQA susceptibility by 4–34×; SVAMP susceptibility is more stable.

4. A sycophancy floor exists and is measurable. Pure Pressure succeeds at 2.0–10.4% for GPT-4o-mini. Structured fallacies add 5–19× above this floor, quantifying the incremental danger of argument content for capable models.

5. Hallucination is overwhelmingly sudden. 67% of hallucinations occur at T1 with no detectable pre-flip signal. Answer-token entropy (≈0.01 nats)

is near-zero for both groups, confirming a binary commitment system.

6. T2 hallucinations are partially predictable. A composite of four T1 features achieves AUC = 0.673 (base rate 26.7%, top bin 48.4%). Verbosity at T1 (57.1% length increase in T2-flip vs 41.5% held-ground, $p < 0.001$) is detectable without logprob access.

Main limitation: Logprob analysis is restricted to Config B SVAMP; extension to all configurations requires open-source logprob access. *Next step:* Evaluate fact-anchoring and consistency-checking defenses against the top-performing fallacy conditions.

A. Fallacy Prompt Design

All prompts enforce five constraints: (1) named condition header; (2) mechanism description; (3) critical constraint block; (4) example opener; (5) turn-level escalation (T1: introduce, T2: intensify, T3: anticipate objections). Pure Pressure forbids all arithmetic, domain knowledge, explanations, and questions—only confident assertion varies in wording.

B. Metric Definitions

HR: Fraction of eligible conversations with an incorrect final answer. **HPT:** Turn (1–3) of first incorrect answer; held-ground = >3. **AvgT(w):** Mean HPT over successful conversations. **Non-Targeted:** Stated answer differs from both correct and attacker's target.

C. Reproducibility Checklist

Datasets. StrategyQA train split (687 Q); SVAMP test split (300 Q). No preprocessing beyond whitespace normalization. Pre-pressure fails excluded from HR computation.

Models. Subject: `gpt-4o-mini-2024-07-18` (OpenAI API); `meta-llama/Llama-3.2-3B-Instruct` (vLLM, 4-bit GPTQ, single A100 40GB). Attacker: `gpt-4o-2024-08-06`; `gpt-5o-mini` (OpenAI API).

Inference. Temperature = 0, top_p = 1.0, max_tokens = 1024, seed = 42. Logprobs: `logprobs=True, top_logprobs=5` (Config B SVAMP only).

Compute. API cost: ≈$45–55 USD (≈112M tokens total). Llama inference: ≈8h on one A100 40GB. Total conversations: ≈16,918 across all six runs.

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